

Souri Rishik Volety

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EDUCATION

Manipal Institute of Technology <i>Bachelor of Technology, Department of Data Science Engineering</i>	Manipal, Karnataka Jul 2023 - Sep 2027
Mahatma Gandhi Memorial PU College, Karnataka State Board <i>Completed Higher Secondary Education in PCMC</i>	Udupi, Karnataka Jul 2021 - Apr 2023

RELEVANT COURSEWORK

- Data Structures & Algorithms
- Database Systems
- Computer Networks
- Machine Learning
- Operating Systems
- Deep Learning
- Data Analysis
- Probability & Statistics

EXPERIENCE

Coding Subsystem Head <i>Robomanipal</i>	Sep 2025 - Present <i>Manipal Institute of Technology</i>
• Revamped and optimized the official Robomanipal website (<i>robomanipal.com</i>) boosting engagement with a 100%+ growth in traffic, and a 98/100 Real Experience Score, ensuring fast load times, robust UX. • Represented MIT MAHE at Technoxian 8.0 (Noida), competing in Line Follower and Maze Solver, and advanced to quarterfinal stage among 200+ teams, demonstrating embedded systems expertise. • Secured 2nd Runner-Up at Robonautica, IISc Bangalore among 179 teams nationwide, for developing a leader-follower robot using ArUco marker-based tracking to maintain distance, follow paths, and avoid collisions.	
Research Intern <i>Manipal Institute of Technology</i>	June 2025 - July 2025 <i>Manipal, Karnataka</i>
• Engineered multiple machine learning models (Gradient Boosting, CatBoost, XGBoost, Random Forest) to detect phishing URLs, achieving up to 97.4% accuracy and 97.7% F1-score on 30,000+ samples. • Performed feature extraction on lexical, structural, and content-based URL attributes, improving F1-score by 12% compared to baseline models. • Designed and deployed a Python-based phishing detection web application with real-time URL classification, reducing manual verification time by 70%.	

PROJECTS

Sign Language Gesture Analysis , <i>Python, TensorFlow, Keras, OpenCV, Numpy, Pandas</i>	Oct 2025 - Nov 2025
• Built a lightweight custom CNN (~227K parameters) to classify 24 static ASL hand gestures from the Sign Language MNIST dataset using normalization, augmentation, and regularization techniques. • Achieved 99.69% validation accuracy and demonstrated real-time webcam inference, correctly recognizing 20/24 gestures under live conditions.	
Digit Recognition , <i>Python, TensorFlow, Keras, Numpy, Pandas</i>	Mar 2025 - Apr 2025
• Developed a Feedforward Neural Network (FNN) in TensorFlow/Keras with preprocessing, batch normalization, dropout, and L2 regularization for MNIST digit recognition. • Achieved 98.13% classification accuracy, validating the effectiveness of optimized FNN architectures.	
Fastest Line Follower , <i>C, Python, STM32CubeIDE</i>	Jul 2024 - Sep 2024
• Optimized PID algorithm with manual, Ziegler-Nichols, and incremental tuning, improving motor precision by 30% at high speeds and reducing oscillations for stable real-time control. • Created a Bluetooth-based Python GUI for real-time PID tuning, cutting adjustment time by 50%, enabling live monitoring, and logging performance data for analysis.	
Open Volley , <i>Python, OpenCV</i>	Jan 2024 - Feb 2024
• Built a computer vision system in Python/OpenCV to track volleyball trajectories with 95% detection accuracy, processing 120+ frames per second for real-time analysis. • Automated ball tracking and player count per team, reducing manual match annotation time by 60% and improving analysis accuracy.	

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL (Oracle), TypeScript/JavaScript, HTML/CSS
Frameworks: React.js/Next.js, NumPy, Pandas, Keras, Scipy, Pytorch, Matplotlib, TensorFlow, OpenCV, CUDA, ROS/ROS2
Developer Tools: Git, GitHub Actions, AWS, VS Code, STM32CubeIDE, Ubuntu
Certifications: Generative AI with Large Language Models, Natural Language Processing in TensorFlow